Pneumonia Detection Using Anchor-Free vs. Anchor-Based Object Detection

Reproducing and Extending Wu et al. (2024)

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Outline

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3. Detection paradigms & models
4. Training recipe & numerical analysis
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Take-home message: the contribution is ***methodological****, a fair, statistically honest comparison, not a new state-of-the-art model.*

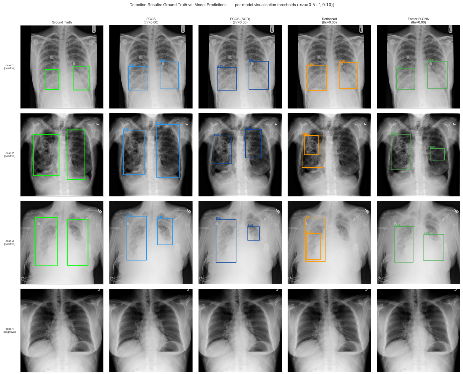
Problem & Dataset

Task: object detection, not classification

- Not "is there pneumonia?" but **where**, predict bounding boxes + confidence.
- Opacities are subtle, variable in shape, overlapping.

RSNA Pneumonia Detection Challenge

- **26,684** chest X-rays (DICOM), box annotations.
- Only $\sim$**22%** positive $\Rightarrow$ strong class imbalance.
- Patient-level 80/20 split (seed 42), no patient leakage.



Green = ground truth; colours = per-model predictions.

The Reference Paper (Wu et al. 2024) & Our Question

What they did: an **anchor-free** detector (FPN + two-branch head + focal loss), trained with **Adam** (lr 10^{-3}, batch 1) on a single RTX 2080ti.

Their reported AP@0.5 (Table 3):

Method	AP@0.5
SSD	25.2
Faster R-CNN	36.5
RetinaNet	45.8
FCOS	48.6
Proposed (anchor-free)	**51.5**

Their claim: anchor-free $>$ anchor-based.

Our research question

But they compare against baselines run with *their own* pipeline (no code released). **At a shared backbone, recipe and evaluation, who really wins?**

Two Paradigms, One Backbone

Anchor-based

- Predefined anchor boxes = a geometric prior on shapes.
- Classify + regress offsets from each anchor.
- **RetinaNet** (1-stage, focal loss), **Faster R-CNN** (2-stage, RPN + ROI).
- Cost: hand-tuned anchor scales/ratios.

Anchor-free (FCOS)

- Per pixel: class score, 4 edge distances (l, t, r, b), a *center-ness*.
- No anchor hyperparameters.
- Ranking score $= p \cdot$ center-ness.

Shared configuration (the experimental control)

All detectors: **ResNet-50 + FPN** backbone, 512×512 input, identical evaluation. Differences therefore come from the **paradigm**, not from feature-extraction capacity.

Training Recipe

A modern recipe (which the paper does not use) applied *identically* to all detectors:

- **Adam**, lr 10^{-3}, 40 epochs.
- **Cosine** annealing + 2-epoch warm-up.
- **EMA** of weights ($\mu = 0.999$).
- BF16 mixed precision (4×H100).
- Weighted sampling of positive patients (3×).
- 3-epoch backbone freeze.
- Medical augmentation (CLAHE, affine. . .).
- Eval: horizontal-flip **TTA** + **Soft-NMS**.

Optimiser note

The paper uses Adam, so **FCOS (Adam)** is the optimiser-faithful run. We add **FCOS (SGD)** as *our own* ablation. Total cost: ∼3.5 h, ∼$30–40.

Numerical Analysis, the NAML core (1/2)

EMA of weights = a linear low-pass (FIR) filter

$\bar{\theta}_t = (1-\mu)\sum_k \mu^{\,t-k}\theta_k$: positive weights summing to 1. Time constant $T_{\text{eff}} = \frac{1}{1-\mu} = 1000$ steps ≈ 1.5 epochs $\Rightarrow$ attenuates high-frequency gradient noise $\Rightarrow$ smoother validation.

Focal loss = a power-law down-weighting of easy examples

$\left|\partial L_{\text{FL}}/\partial p\right| \sim (1-p)^{\gamma}(1+\gamma)$. With $\gamma = 2$, an easy positive at $p = 0.9$ is weighted $\sim 36\times$ less than a hard one.

GIoU = a differentiable surrogate for IoU

IoU has *zero* gradient for disjoint boxes; GIoU adds an enclosing-box term whose gradient is non-zero everywhere $\Rightarrow$ a usable regression loss (no extra hyperparameters).

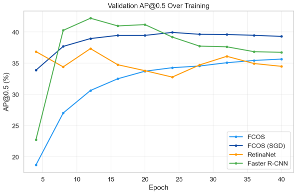
Numerical Analysis, the NAML core (2/2)

Cosine annealing with warm-up

$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos(\pi t/T))$, with $\eta'(0) = \eta'(T) = 0$: **smooth start and end** (no step-decay shock). Warm-up counters the zero-biased Adam moments $\hat{m}, \hat{v}$ in the first steps.

Bootstrap confidence intervals: Monte-Carlo convergence

Quantile MC error decays as $O(1/\sqrt{B})$: $\mathrm{SE}_{0.975} \approx \frac{1}{f(q)}\sqrt{\frac{p(1-p)}{B}} \approx 0.19\,\text{AP}$ at $B = 200$, well below our $\sim$1 AP resolution. CI *width* is set by patient variance, not by B.



Metrics

Detection

- **AP@0.5** (101-point, COCO): headline.
- Size-stratified AP on *RSNA* tertiles (COCO buckets are degenerate here).
- AR_{10}: recall at 10 detections/image.

Patient-level

- F1 at threshold τ (paper uses $\tau = 0.3$).
- **ROC AUC**: threshold-free, calibration-insensitive.
- **FROC/CPM**, **ECE** (calibration).

Why so many metrics?

Because a single fixed-threshold number can be dominated by **score calibration** rather than detection quality, the central subtlety of our results.

Results: Detection Performance

Method	AP@0.5 [95% CI]	ROC AUC	note
FCOS (Adam)	35.3 [32.6, 38.4]	86.5	optimiser-faithful
FCOS (SGD, our ablation)	39.8 [37.1, 43.1]	88.1	
RetinaNet	41.2 [38.2, 44.1]	89.1	
Faster R-CNN	**42.9** [40.0, 46.3]	88.9	**our best**
Ensemble (FCOS+Retina, WBF)	42.0 [39.2, 45.6]	87.5	
Wu et al. FCOS	*48.6*	–	*their eval*
Wu et al. proposed	*51.5*	–	*their eval*

- In *our* controlled setup: **anchor-based** > **anchor-free** (Faster R-CNN, RetinaNet top FCOS), the **reverse** of the paper's ordering.
- ROC AUC lies in a **tight 86.5–89.1%** band: true ranking quality is similar.

The Honest Paper Comparison

We do NOT beat the paper's absolute numbers, and we say so

Our best is 42.9; the paper reports FCOS 48.6 and 51.5 for its method.

Why? An evaluation gap, not a training gap:

- Our recipe is if anything *stronger* (EMA, mixed precision, medical augmentation).
- But the paper's recall is far higher: $\mathrm{AR}_{10} \approx$ **96–99** vs. our $\approx$ **85**. Higher recall at fixed IoU **inflates AP**, regardless of paradigm.
- No code, weights or split released $\Rightarrow$ protocols cannot be aligned.

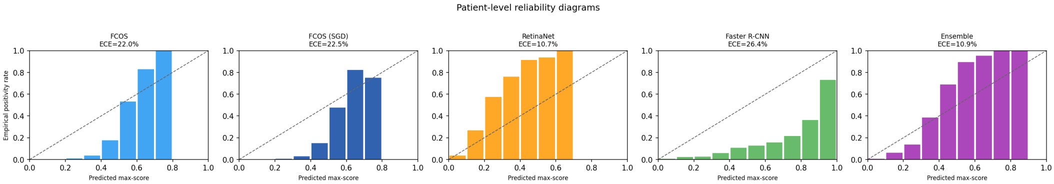
The defensible claim

The only protocol-matched comparison is the **internal** one, and there the paradigm ordering **reverses** relative to the paper. That is our contribution.

The Calibration Artefact (our strongest result)

At fixed $\tau = 0.3$, the WBF ensemble "wins" patient F1 (63.0%, +9.8 over the best single). **But this is a calibration artefact:**

- Detectors' optimal thresholds differ wildly (FCOS 0.47, RetinaNet 0.13, F-RCNN 0.82); a fixed cut penalises **mis-calibrated** models, not worse models.
- Under a *fair* operating point (Youden on a held-out half, or a learnt 5-fold aggregator) the ensemble edge shrinks to ≤ 1 F1 and is **overtaken by RetinaNet** (learnt-aggregator F1 **67.3%**, best in the study).



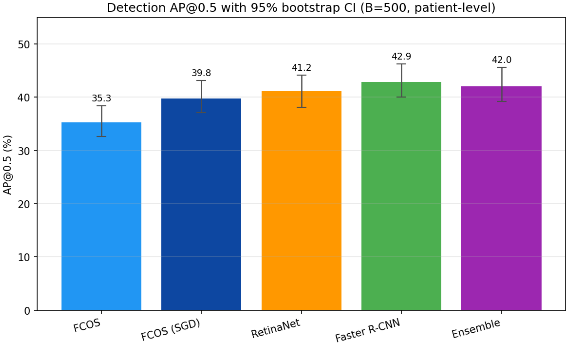
Reliability diagrams (predicted confidence vs. empirical positivity); ECE above each panel.

Robustness Checks (no extra training)

All run on cached predictions:

- Patient-level **bootstrap CIs** on AP@0.5.
- Three-way split for the threshold.
- RSNA-percentile size buckets.
- FROC/CPM, reliability/ECE.
- Learnt 5-fold patient aggregator.
- **Paired bootstraps** on AP differences.

Only Faster R-CNN strictly beats another detector (-3.07 vs. FCOS-SGD, $p < 0.001$); the ensemble does *not* beat RetinaNet ($+0.88$, $p = 0.25$).



Limitations (stated up front)

- **Single training seed** per detector: run-to-run variance ($\pm$1–2 AP) is not captured $\Rightarrow$ we treat AP differences $<$2 points as unresolved.
- **No held-out test set**: the same split is used for selection, thresholds and CIs (selection-on-val bias).
- **Asymmetric ablation**: SGD tried only on FCOS $\Rightarrow$ paradigm vs. optimiser separated on the FCOS side only.
- No per-site stratification, no external dataset; cross-paper AP not directly comparable.

Disclosing these *before* being asked is the point: rigour and honesty over a polished-but-fragile headline.

Conclusion

1. Under a **shared backbone, recipe and evaluation**, anchor-based detectors beat anchor-free FCOS on AP@0.5, the **reverse** of the paper's paradigm claim.
2. We do *not* reproduce the paper's absolute numbers; the gap is one of **evaluation** (recall $\sim$96–98 vs. 85), not training.
3. The ensemble's patient-level "win" is a **calibration artefact**: at a fair operating point RetinaNet is the best single model (F1 67.3%).
4. The numerical-analysis backbone (EMA filter, focal gradient, GIoU, cosine, bootstrap) justifies every training choice.

The contribution is a fair, honest, statistically grounded comparison.

Thank you

Questions?

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Code: `github.com/moa155/pneumonia-detection-naml`